

CBSE 2027 Maths — Predicted Paper (Set 7 of 10)

Based on the official blueprint and the highest-weightage topics identified in the latest CBSE Class 12 trends, here is Set 7 of the predicted board paper. This paper integrates the most highly repeated PYQ formats and competency-based questions.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $f(x) = [x]$ (greatest integer function) and $g(x) = |x|$. Find the value of $g(f(-5/3)) - f(g(-5/3))$. * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q2. Find the exact value of $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If $A = \begin{pmatrix} 0 & 2 \\ 3 & -4 \end{pmatrix}$ and $kA = \begin{pmatrix} 0 & 3a \\ 2b & 24 \end{pmatrix}$, find the scalar value k . * **Chapter:** Matrices * **Probability Tag: High**

Q4. If A is a square matrix of order 3 such that $A(\text{adj } A) = 10I$, find the value of $|A|$. * **Chapter:** Determinants * **Probability Tag: Very High**

Q5. What is the derivative of $\log_e(\cos e^x)$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The function $y = x^2 e^{-x}$ is strictly increasing in which interval? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate the indefinite integral: $\int e^x \left(\frac{x-1}{x^2}\right) dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q8. What is the area of the region bounded by the curve $y = \sin x$, the x-axis, and the ordinates $x = 0$ and $x = \frac{\pi}{2}$? * **Chapter:** Application of Integrals * **Probability Tag: High**

Q9. Determine the order and degree (if defined) of the differential equation: $\left(\frac{ds}{dt}\right)^4 + 3s \frac{d^2s}{dt^2} = 0$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q10. Find the unit vector in the direction of the sum of the vectors $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q11. Find the area of a triangle whose adjacent sides are represented by the vectors $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = -3\hat{i} + 7\hat{j}$. * **Chapter:** Vector Algebra * **Probability Tag: Medium**

Q12. Find the intercept cut off by the plane $2x - 3y + 4z = 12$ on the x-axis. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q13. Write the Cartesian equation of the line passing through $(1, 2, -4)$ and parallel to the vector $\hat{i} + 2\hat{j} - \hat{k}$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q14. In an LPP, if the objective function $Z = ax + by$ attains its same maximum value at two distinct corner points of the feasible region, then the number of points at which Z is

maximum is: * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. If $P(A) = 0.6$, $P(B) = 0.3$, and $P(A \cap B) = 0.2$, find $P(A|B')$. * **Chapter:** Probability * **Probability Tag: High**

Q16. Evaluate the definite integral: $\int_{-1}^1 \log\left(\frac{2-x}{2+x}\right) dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$. * **Chapter:** Differential Equations * **Probability Tag: Medium**

Q18. A bag contains 3 red and 5 black balls. If two balls are drawn at random without replacement, what is the exact probability that both are red? * **Chapter:** Probability * **Probability Tag: High**

Q19. (Assertion-Reason): Assertion (A): For any two non-singular matrices A and B , $(AB)^{-1} = B^{-1}A^{-1}$. **Reason (R):** Matrix multiplication is generally not commutative. * **Chapter:** Matrices * **Probability Tag: Very High**

Q20. (Assertion-Reason): Assertion (A): For any vector $\vec{a}$, $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2\vec{a}$. **Reason (R):** Any vector $\vec{a}$ in space can be written as $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Solve for x : $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q22. Find $\frac{dy}{dx}$ if $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, where $0 < x < 1$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q23. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} - 2\hat{j} + 3\hat{k}$, and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$, find a vector of magnitude 6 units which is parallel to the vector $2\vec{a} - \vec{b} + 3\vec{c}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q24. Solve the differential equation by rearranging terms: $(x + y) \frac{dy}{dx} = 1$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. A die is thrown once. If E is the event "the number appearing is a multiple of 3" and F is the event "the number appearing is even," find whether E and F are mathematically independent events. * **Chapter:** Probability * **Probability Tag: Medium**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R in the set of integers $\mathbb{Z}$ given by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation. * **Chapter:** Relations and Functions * **Probability Tag: Very High**

Q27. Find the matrix X such that $X \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 0 & 1 \end{pmatrix}$. * **Chapter:** Matrices * **Probability**

Tag: High

Q28. Prove that the curves $x = y^2$ and $xy = k$ intersect orthogonally (cut at right angles) if $8k^2 = 1$. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q29. Evaluate the definite integral: $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the homogeneous differential equation: $x \frac{dy}{dx} = y - x \tan\left(\frac{y}{x}\right)$. * **Chapter:** Differential Equations * **Probability Tag: Very High**

Q31. Three machines A , B , and C produce 50%, 30%, and 20% of the items in a factory, respectively. The percentage of defective outputs are 3%, 4%, and 5% respectively. If an item is drawn at random from the day's production and found defective, find the probability it was produced by machine A . * **Chapter:** Probability * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Solve the following system of equations using the matrix inverse method:
 $\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$ $\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$ $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$ * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q33. Show that the altitude of the right circular cylinder of maximum volume that can be inscribed in a given sphere of radius R is $\frac{2R}{\sqrt{3}}$. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Using integration, find the area of the region defined by:
 $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q35. Find the vector and Cartesian equations of the line passing through the point $(1, -1, 1)$ and perpendicular to the lines joining the points $(4, 3, 2)$, $(1, -1, 0)$ and $(1, 2, -1)$, $(2, 1, 1)$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Investment Strategy) A trust fund has ₹30,000 that must be invested in two different types of government bonds. The first bond pays 5% interest per year, and the second bond pays 7% interest per year. The trust wishes to use matrix multiplication to determine how to divide the money. (i) Formulate the investment amounts and the interest rates as two matrices. (1 Mark) (ii) Write the matrix equation to find the total annual interest. (1 Mark) (iii) Solve the matrix equation to determine exactly how to divide the ₹30,000 among the two types of bonds if the trust fund must obtain an annual total interest of exactly ₹1800. (2 Marks) * **Chapter:** Matrices * **Probability Tag: High**

Q37. Case Study 2: Application of Derivatives (Publishing Layout) A publishing company is printing a poster. The printed page is required to have a total fixed area of **24 sq inches**. The margins at the top and bottom are **1.5 inches** each, and the margins at the left and right sides are **1 inch** each. Let x be the overall width of the page and y be the overall height of the page. (i) Express the actual printed area A inside the margins as a function of the page width x . (1 Mark) (ii) Find $\frac{dA}{dx}$ and equate it to zero to find the critical point. (1 Mark) (iii) Use the second derivative test to determine the overall dimensions of the page (x and y) so that the printed area is maximized. (2 Marks) * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Carnival Dice Game) A game at a school fair involves throwing a standard pair of dice. To play, a student pays an entry fee. The rules dictate that the student wins ₹15 if the sum of the numbers on the dice is 7, wins ₹10 if the sum is 11, and loses ₹5 for any other sum. Let X be the random variable representing the amount won or lost by the student. (i) What is the exact probability of rolling a sum of 7 and winning ₹15? (1 Mark) (ii) What is the probability of losing ₹5? (1 Mark) (iii) Create the probability distribution table for X and find the expected value (mean) of the amount won or lost by the student per game. (2 Marks) * **Chapter:** Probability * **Probability Tag: High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.